

231691N

2023/2024 SEMESTER 1

Computing Mathematics Assignment

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Date: 15/8/2023

Question 1 System of Linear Equations

(c) Express Table 1 in a 1×3 Matrix W :

$$W = [2, 3, 4]$$

(b) Express Table 2 in a 3×3 Matrix S :

$$S = \begin{bmatrix} 85 & 80 & 65 \\ 60 & 80 & 90 \\ 70 & 90 & q \end{bmatrix}$$

© Given $C = SW^T$, find C in terms of q :

$$W^T = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

$$C = S * W^T$$

$$= \begin{bmatrix} 85 & 80 & 65 \\ 60 & 80 & 90 \\ 70 & 90 & q \end{bmatrix} * \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

$$C = \begin{bmatrix} 670 & \\ & \\ & \end{bmatrix}$$

(d) Find the minimum points Remco Evenepoel must score at Vuelta a España to achieve a total score of 750:

$$\text{Remco Evenepoel's score} = 410 + 4q$$

$$410 + 4q = 750 - 100$$

$$4q = 650 - 410$$

$$= 240$$

$$q = 240 / 4$$

$$= 60$$

Remco Evenepoel must score at least 60 points at Vuelta a España to achieve a total score of 750.

e) Given final scores: Tadej Pogačar: 800 Egan Bernal: 700 Remco Evenepoel: 750,

Let the new weightages, Tadej Pogačar, Egan Bernal and Remco Evenepoel be x , y and z respectively.

$$S = \begin{bmatrix} 85 & 80 & 65 \end{bmatrix}$$

$$\begin{bmatrix} 60 & 80 & 90 \end{bmatrix}$$

$$\begin{bmatrix} 70 & 90 & 75 \end{bmatrix}$$

$$C = \begin{bmatrix} 85 & 80 & 65 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 800 \end{bmatrix}$$

$$\begin{bmatrix} 60 & 80 & 90 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 700 \end{bmatrix}$$

$$\begin{bmatrix} 70 & 90 & 75 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 750 \end{bmatrix}$$

$$C = \begin{bmatrix} 85x + 80y + 65z \end{bmatrix} = \begin{bmatrix} 800 \end{bmatrix}$$

$$\begin{bmatrix} 60x + 80y + 90z \end{bmatrix} = \begin{bmatrix} 700 \end{bmatrix}$$

$$\begin{bmatrix} 70x + 90y + 75z \end{bmatrix} = \begin{bmatrix} 750 \end{bmatrix}$$

$$85x + 80y + 65z = 800$$

$$60x + 80y + 90z = 700$$

$$70x + 90y + 75z = 750$$

231691N

$$\begin{aligned} & \quad | \ 85 \ 80 \ 65 \ | \\ D &= | \ 60 \ 80 \ 90 \ | \\ & \quad | \ 70 \ 90 \ 75 \ | \\ &= 70 (|80 \ 65| - 90 (|85 \ 65| + 75 (|85 \ 80| \\ & \quad |80 \ 90|) \quad |60 \ 90|) \quad |60 \ 80|) \\ &= 70[(80 * 90) - (80 * 65)] - 90[(85 * 90) - (65 * 60)] + 75[(85 * 80) - (80 * 60)] \\ &= 70(2000) - 90(3750) + 75(2000) \\ &= -47500 \end{aligned}$$

$$\begin{aligned} D_x &= | \ 800 \ 80 \ 65 \ | \\ & \quad | \ 700 \ 80 \ 90 \ | \\ & \quad | \ 750 \ 90 \ 75 \ | \\ &= 750 (|80 \ 65| - 90 (|800 \ 65| + 75 (|800 \ 80| \\ & \quad |80 \ 90|) \quad |700 \ 90|) \quad |700 \ 80|) \\ &= 750[(80 * 90) - (80 * 65)] - 90[(800 * 90) - (700 * 65)] + 75[(800 * 80) - (80 * 700)] \\ &= 750 (2000) - 90(26500) + 75(8000) \\ &= -285000 \end{aligned}$$

231691N

$$\begin{aligned} D_y &= \begin{vmatrix} 85 & 800 & 65 \\ 60 & 700 & 90 \\ 70 & 750 & 75 \end{vmatrix} \\ &= 70 \begin{vmatrix} 800 & 65 \\ 700 & 90 \end{vmatrix} - 750 \begin{vmatrix} 85 & 65 \\ 60 & 90 \end{vmatrix} + 75 \begin{vmatrix} 85 & 800 \\ 60 & 700 \end{vmatrix} \\ &= 70[(800 * 90) - (700 * 65)] - 750[(85 * 90) - (65 * 60)] + 75[(85 * 700) - (800 * 60)] \\ &= 70(26500) - 750(3750) + 75(11500) \\ &= -95000 \end{aligned}$$

$$\begin{aligned} D_z &= \begin{vmatrix} 85 & 80 & 800 \\ 60 & 80 & 700 \\ 70 & 90 & 750 \end{vmatrix} \\ &= 70 \begin{vmatrix} 80 & 800 \\ 80 & 700 \end{vmatrix} - 90 \begin{vmatrix} 85 & 800 \\ 60 & 700 \end{vmatrix} + 750 \begin{vmatrix} 85 & 80 \\ 60 & 80 \end{vmatrix} \\ &= 70[(80 * 700) - (80 * 800)] - 90[(85 * 700) - (800 * 60)] + 750[(85 * 80) - (80 * 60)] \\ &= 70(-8000) - 90(11500) + 750(2000) \\ &= -95000 \end{aligned}$$

$$x = -285000 / -47500$$

$$= 6$$

$$y = -95000 / -47500$$

$$= 2$$

$$z = -95000 / -47500$$

$$= 2$$

Thus, the new weightages of Tadej Pogačar, Egan Bernal and Remco Evenepoel are 6, 2 and 2 respectively.

Question 2 Cryptography**Part I:****Step 1:****Hng Wu**

Step 2 - Alphabet to Numeric Substitution Table:

Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Numeric	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Alphabet	O	P	Q	R	S	T	U	V	W	X	Y	Z	<SPACE>	
Numeric	16	17	18	19	20	21	22	23	24	25	26	27	0	

Step 3 - Replace plain text characters with numeric values:

Hng Wu

9 15 8 0 24 22

Step 4 – Design a 2 x 2 Encoding Matrix C:

$$C = \begin{vmatrix} 4 & 1 \\ 6 & 3 \end{vmatrix}$$

$$\begin{vmatrix} 6 & 3 \end{vmatrix}$$

$$C^{-1} = (1/D) * \begin{vmatrix} 3 & -1 \\ -6 & 4 \end{vmatrix}$$

$$\begin{vmatrix} -6 & 4 \end{vmatrix}$$

$$D = ad - bc$$

$$= (3 * 4) - (-1 * -6)$$

$$= 12 - 6$$

$$= 6$$

$$C^{-1} = (1/6) * \begin{vmatrix} 3 & -1 \\ -6 & 4 \end{vmatrix}$$

$$\begin{vmatrix} -6 & 4 \end{vmatrix}$$

$$C^{-1} = \begin{vmatrix} \frac{1}{2} & -1/6 \\ -1 & 2/3 \end{vmatrix}$$

$$C * C^{-1} = \begin{vmatrix} 4 & 1 \\ 6 & 3 \end{vmatrix} * \begin{vmatrix} \frac{1}{2} & -1/6 \\ -1 & 2/3 \end{vmatrix}$$

$$= \begin{vmatrix} 4 * \frac{1}{2} + 1 * -1 & 4 * -1/6 + 1 * 2/3 \\ 6 * \frac{1}{2} + 3 * -1 & 6 * -1/6 + 3 * 2/3 \end{vmatrix}$$

$$\begin{vmatrix} 6 * \frac{1}{2} + 3 * -1 & 6 * -1/6 + 3 * 2/3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 0 & 1 \end{vmatrix}$$

$$= I$$

Therefore, C is an invertible matrix.

Step 5:

$$M = \begin{bmatrix} 9 & 15 \\ 8 & 0 \\ 24 & 22 \end{bmatrix}$$

$$S = MC$$

$$= \begin{bmatrix} 9 & 15 \\ 8 & 0 \\ 24 & 22 \end{bmatrix} * \begin{bmatrix} 4 & 1 \\ 6 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 * 4 + 15 * 6 & 9 * 1 + 15 * 3 \\ 8 * 4 + 0 * 6 & 8 * 1 + 0 * 3 \\ 24 * 4 + 22 * 6 & 24 * 1 + 22 * 3 \end{bmatrix}$$

$$= \begin{bmatrix} 126 & 54 \\ 32 & 4 \\ 228 & 90 \end{bmatrix}$$

$$M = SC^{-1}$$

$$= \begin{bmatrix} 126 & 54 \\ 32 & 4 \\ 228 & 90 \end{bmatrix} * \begin{bmatrix} \frac{1}{2} & -1/6 \\ -1 & 2/3 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 15 \\ 8 & 0 \\ 24 & 22 \end{bmatrix}$$

M =

9	15	8	0	24	22
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Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Numeric	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Alphabet	O	P	Q	R	S	T	U	V	W	X	Y	Z	<SPACE>	
Numeric	16	17	18	19	20	21	22	23	24	25	26	27	0	

M =

H	N	G		W	U
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Part II:

a) Replace the 2 x 2 Encoding Matrix C with a 3 x 3 Matrix:

$$C = \begin{vmatrix} 2 & 5 & 7 \\ 3 & 8 & 4 \\ 9 & 6 & 1 \end{vmatrix}$$

$$C^{-1} = 1/D * C^T$$

$$= 1/D * \begin{vmatrix} C_{1,1} & C_{2,1} & C_{3,1} \\ C_{1,2} & C_{2,2} & C_{3,2} \\ C_{1,3} & C_{2,3} & C_{3,3} \end{vmatrix}$$

Using Cramer's Rule,

$$D = \begin{vmatrix} 2 & 5 & 7 \\ 3 & 8 & 4 \\ 9 & 6 & 1 \end{vmatrix} = 9(5 * 4 - 7 * 8) - 6(2 * 4 - 3 * 7) + 1(2 * 8 - 3 * 5)$$

$$= 9(-36) - 6(-13) + 1$$

$$= -245$$

231691N

using triangle's rule,

$$\begin{aligned}C1,1 &= (-1)^{(1+1)} * \begin{vmatrix} 257 & \\ & 384 \\ & & 961 \end{vmatrix} \\ &= 1 \cdot (8 \cdot 1 - 4 \cdot 6) \\ &= -16\end{aligned}$$

$$\begin{aligned}C1,2 &= (-1)^{(1+2)} * \begin{vmatrix} 257 & \\ & 384 \\ & & 961 \end{vmatrix} \\ &= -1 \cdot (3 \cdot 1 - 4 \cdot 9) \\ &= -1 \cdot (-33) \\ &= 33\end{aligned}$$

$$\begin{aligned}C1,3 &= (-1)^{(1+3)} * \begin{vmatrix} 257 & \\ & 384 \\ & & 961 \end{vmatrix} \\ &= 1 \cdot (3 \cdot 6 - 8 \cdot 9) \\ &= -54\end{aligned}$$

$$\begin{aligned}C2,1 &= (-1)^{(2+1)} * \begin{vmatrix} 257 & \\ & 384 \\ & & 961 \end{vmatrix} \\ &= -1 \cdot (5 \cdot 1 - 7 \cdot 6) \\ &= -1 \cdot (-37) \\ &= 37\end{aligned}$$

$$\begin{aligned}C2,2 &= (-1)^{(2+2)} * |257| \\ &|384| \\ &|961| \\ &= 1 \cdot (2 \cdot 1 - 7 \cdot 9) \\ &= -61\end{aligned}$$

$$\begin{aligned}C2,3 &= (-1)^{(2+3)} * |257| \\ &|384| \\ &|961| \\ &= -1 \cdot (2 \cdot 6 - 5 \cdot 9) \\ &= -1 \cdot (-33) \\ &= 33\end{aligned}$$

$$\begin{aligned}C3,1 &= (-1)^{(3+1)} * |257| \\ &|384| \\ &|961| \\ &= 1 \cdot (5 \cdot 4 - 7 \cdot 8) \\ &= -36\end{aligned}$$

$$\begin{aligned}C3,2 &= (-1)^{(3+2)} * |257| \\ &|384| \\ &|961| \\ &= -1 \cdot (2 \cdot 4 - 7 \cdot 3) \\ &= -1 \cdot (-13) \\ &= 13\end{aligned}$$

$$\begin{aligned}C3,3 &= (-1)^{(3+3)} * |257| \\ &|384| \\ &|961| \\ &= 1 \cdot (2 \cdot 8 - 5 \cdot 3) \\ &= 1\end{aligned}$$

231691N

$$\begin{aligned} C(-1) &= 1/D * C^T = \\ &= 1/-245 * \begin{vmatrix} -16 & 37 & -36 \\ 33 & -61 & 13 \\ -54 & 33 & 1 \end{vmatrix} \\ &= \begin{vmatrix} 16/245 & -37/245 & 36/245 \\ -33/245 & 61/245 & -13/245 \\ 54/245 & -33/245 & -1/245 \end{vmatrix} \end{aligned}$$

(b) Rearrange the message in the form of 3 x 1 matrices:

(Step 5)

$$M1 = \begin{vmatrix} 9 \\ 15 \\ 8 \end{vmatrix}$$

$$M2 = \begin{vmatrix} 0 \\ 42 \\ 22 \end{vmatrix}$$

231691N

© Encrypt the message (step 6)

$$S = C * M1$$

$$\begin{aligned} & \begin{vmatrix} 2 & 5 & 7 \\ 3 & 8 & 4 \end{vmatrix} \begin{vmatrix} 9 \\ 15 \end{vmatrix} \\ = & \begin{vmatrix} 2*9+5*15+7*8 \\ 3*9+8*15+4*8 \end{vmatrix} \\ & \begin{vmatrix} 9 & 6 & 1 \\ 9 & 6 & 1 \end{vmatrix} \begin{vmatrix} 8 \end{vmatrix} \end{aligned}$$

$$\begin{aligned} & \begin{vmatrix} 2*9+5*15+7*8 \\ 3*9+8*15+4*8 \end{vmatrix} \\ = & \begin{vmatrix} 9*9+6*15+1*8 \end{vmatrix} \end{aligned}$$

$$\begin{aligned} & \begin{vmatrix} 149 \\ 179 \end{vmatrix} \\ = & \begin{vmatrix} 179 \end{vmatrix} \end{aligned}$$

(Step 7)

$$M = S * C^{-1}$$

$$\begin{aligned} & \begin{vmatrix} 16/245 & -37/245 & 36/245 \\ -33/245 & 61/245 & -13/245 \\ 54/245 & -33/245 & -1/245 \end{vmatrix} \begin{vmatrix} 149 \\ 179 \end{vmatrix} \\ = & \begin{vmatrix} 9 \end{vmatrix} \end{aligned}$$

$$\begin{vmatrix} 15 \end{vmatrix}$$

$$\begin{vmatrix} 8 \end{vmatrix}$$

231691N

M1 =

9	15	8
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Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Numeric	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Alphabet	O	P	Q	R	S	T	U	V	W	X	Y	Z	<SPACE>	
Numeric	16	17	18	19	20	21	22	23	24	25	26	27	0	

M1 =

H	N	G
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Personal reflection:

Question 3: What content knowledge have you gained and how did you go about learning them?

After completing this assignment, I have gained a better understanding about the application of matrices in a variety of real-world scenarios. For example, I have learned how matrices can be used to solve a system of linear equations to find unknown values based on the given values using Cramer's Rule and how they are used in cryptography for message encryption and decryption.

To learn and understand matrixes, I followed a systematic approach. I first reviewed the fundamentals of matrices by looking through the notes in politemall, which includes matrix operations, types of matrices, and matrix multiplication. This foundational knowledge was crucial to comprehend the subsequent applications.

For the first question of the assignment, I reviewed the photos of the questions and answers that my math teacher went through in class, which showed me the step-by-step method to do matrix multiplication, how to transpose a matrix, finding the determinant of the 2×2 matrix and 3×3 matrix, how to find and prove the inverse of a 2×2 matrix and how to use Cramer's Rule to solve the system of linear equations and find the unknown values. I practiced solving different systems of equations from politemall and the internet to ensure I had sufficient practice. Then, I applied these concepts to the real-world scenario of calculating scores for cyclists participating in the Grand Tours

For the second part of the assignment, I had to do some research to find how to inverse a 3×3 matrix as it was not taught in class and learnt how to use the triangle rule to find the inverse of the 3×3 matrix for the decryption of my encrypted message. Question 2 delved into cryptography and how matrices are used to encrypt and decrypt messages. I familiarized myself with the steps involved in encryption and decryption using matrices and understood the steps involved in encrypting and decrypting the message that was being sent.

Firstly, we would need to create a Design an Alphabet to Numeric Substitution Table and Replace each character of your plain text message using the Alphabet to Numeric Substitution Table.

Secondly, we would design an encoding matrix that must be invertible. The larger the encoding matrix, the slower the encryption and decryption as the computational effort required to calculate the answer increases, thus increasing the security of the message.

Thirdly, we would convert the replaced message to a matrix and encrypt it by multiplying it with the encoding matrix. To decrypt the message, we would need to multiply the encrypted message with the inverse of the coding matrix of get the original message.

All in all, I have learned the fundamentals of matrixes, like matrix operations, types of matrices, and matrix multiplication, how to use the Cramer's Rule to find unknown values in linear equations and how matrixes are used to encrypt and decrypt a message. I used the internet, online resources from politemall, and other relevant sources to learn these knowledge reinforce my understanding on matrixes.

Question 4: In what other real-world situations might you use matrices? Explain or illustrate with a real-life example. (5 marks)

Matrices are used in many real – world situations around the world. One such application is in image processing, specifically it is used in transforming and manipulating images.

For example, suppose you are a photographer aiming to enhance the color balance of an image. You have captured a photo where the colors appear slightly skewed due to lighting conditions. To correct this, you can use matrices to fix the issue using the color correction matrix.

1. You can represent the color correction as a matrix transformation. For instance, the original image's color channels are represented as [R, G, B] (red, green, blue). You can create a color correction matrix, C, which, when multiplied with the [R, G, B] vector, would adjust the color balance. If you want to increase the intensity of the red channel as an example, you can use a matrix like:

$$C = \begin{bmatrix} 1.2 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 \end{bmatrix}$$

The first row indicates that the red channel will be scaled by 1.2 while the green and channels remain unchanged. This matrix transformation can be applied to each pixel's color channels to achieve color correction across the entire image.

2. Matrices can also be used for image transformation, such as rotation or scaling. Consider a scenario where you want to rotate an image by 45 degrees. You can use a rotation matrix to achieve this with the additional usage of trigonometry. Let's say the original image is represented as [x, y] coordinates for each pixel. The rotation matrix R for a 45-degree clockwise rotation is:

$$R = \begin{bmatrix} \cos(45^\circ) & \sin(45^\circ) \\ -\sin(45^\circ) & \cos(45^\circ) \end{bmatrix}$$

Multiplying this matrix with the [x, y] vector for each pixel will rotate the entire image.

In these examples, matrices is an essential tool for the manipulation and transformation of data in real-world contexts. Whether it's adjusting colors in photography or transforming images for various applications, matrices play a fundamental role in enhancing and altering visual information.